

Jakob Shurtleff's CV

Jakob Shurtleff

Computational Physics Student, Programmer, Researcher.

jakobshurtleff@email.com | [My GitHub Profile](#) | [Download PDF](#)

Currently

6th Semester Computational Physics student focusing on semiconductor theory and AI integration.

Specialized in

Computational modeling, Python-based automation, multi-agent AI systems, and physics lab coordination.

Research interests

Quantum mechanics, GAAFET/MOS capacitor simulation, astrophysics, and local data aggregation.

Education

Sept 2022 - Apr 2026 **Brigham Young University**—Idaho, Rexburg, ID

- Bachelor of Science in Physics
- Computational Concentration

Technical Skills

Programming Python (Polars, Scipy, Lets-Plot), C++, Ollama, Web Scraping.

Tools Linux, Git, Quarto, LaTeX, RTX-accelerated simulations.

Languages English (Native), Mandarin Chinese, Thai.

Projects

Artificial Intelligence

2026 **Multi-Agent Chatbot** Developed an interactive AI system using Python and Ollama for complex task delegation and reasoning.

2026 **AI Fish Simulation** Designed and debugged a reward-based AI system for aquatic behavioral simulations in C++.

Data Science

2026 **Local Event Scraper** Built a Python-based web scraper to aggregate and format event data for Idaho Falls and Rexburg newsletters.

Experience

Feb 2026 - Present **Physics Lab Coordinator**, BYU-Idaho

- Managed laboratory equipment and experimental setups.
- Supported faculty in delivering undergraduate physics curriculum.

2025 - 2026 **Independent Developer**, Rexburg, ID

- Optimized hardware-specific environments for RTX 2070 systems.
- Developed and maintained custom Minecraft modpacks on NeoForge.